NullState Phase 1 Protocol: Hardening the Neural Vertical

TL;DR

- **Roughly 60% of Phase 1 is executable today** with the pilot's existing latent space and on-disk objects — classifier hardening, geosketch scaffolding, the geometry/statistics stack (sliced-Wasserstein, Sinkhorn, persistence), Control-3 maturation scoring via CytoTRACE2, contrastiveVI dish-stripping setup, CellOracle levers, and signature anchoring — none of which strictly require new data.
- **The single hard blocker is true raw UMI counts for the HNOCA query.** They are almost certainly retrievable (CELLxGENE-hosted HNOCA carries a schema-mandated raw layer; per-study GEO/ArrayExpress matrices exist, e.g. Pellegrini ChP = E-MTAB-10500), but no Zenodo HNOCA object is *confirmed* to contain an integer-counts layer, so step 1 begins with a download-and-inspect verification, not an assumption.
- **The full protocol fits the USD 150–200 ceiling with comfortable margin** (~$70–120 total; only ~$12–23 of pure compute at risk-adjusted spot/on-demand neo-cloud rates) by training every deep model exactly once on geosketch-enriched subsets, projecting the rest, and keeping all optimal-transport and topology compute off the GPU.

Key Findings

1. **Raw counts are the lever, and they exist — but must be verified, not assumed.** The pilot's NaN divergence and 0-epoch fallback are a direct, mechanical consequence of feeding log-normalized / length-normalized values into a ZINB/NB likelihood that expects integers. This is not a tuning problem; it is a likelihood-misspecification problem with one correct fix. scArches documentation is explicit (verbatim, docs.scarches.org SCVI/SCANVI surgery pipelines): *"This line makes sure that count data is in the adata.X. Remember that count data in adata.X is necessary when using 'nb' or 'zinb' loss."* The trVAE training-tips page adds: *"We recommend you to set recon_loss = nb or zinb. These loss functions require access to count data. You need to have raw count data in adata.raw.X. If you don't have access to count data and have normalized log-transformed data then set recon_loss to mse."* Three HNOCA Zenodo objects exist — cleaned `hnoca_cleanedmeta.h5ad` (18.7 GB, zenodo 14161275), full `hnoca_allmeta.h5ad` (48.9 GB, zenodo 14160929), and a minimal mapping set (zenodo 15004817) — but none is documented to carry a raw integer-counts layer; HNOCA-tools only ever references `layer="lognorm"` and `scvi_normalized`. The CELLxGENE-hosted HNOCA (~216k cells, a subsample) is schema-guaranteed to expose raw counts in `X['raw']`/`adata.raw`. The robust path is therefore per-study raw matrices reconciled by barcode. GitHub GitHub
2. **The classifier fix is trivial and high-yield (~1–2 h GPU, ~$1–3).** scvi-tools exposes `scvi.train._callbacks.SaveCheckpoint` (with `best_model_path`/`best_model_score`, and an `on_exception` option since 1.3.0 that saves the best model on NaN loss/gradient), plus `early_stopping_monitor='elbo_validation'`. scANVI semi-supervised training should additionally monitor classification accuracy (the scArches docs note "the best earlystopping criteria are the 'elbo' for SCVI pretraining ... and 'accuracy' for semi-supervised SCANVI training"). The uniform class prior is the wrong default when organoid query composition differs sharply from the primary reference.
3. **contrastiveVI is the scientific centerpiece and is fully specified in scvi-tools** (`scvi.external.ContrastiveVI`, `representation_kind ∈ {"salient","background"}`, `n_salient_latent`/`n_background_latent`). Its validity hinges entirely on Control 3, because the pilot's 0.906 dish-cosine could be confounded by shared immaturity.
4. **The maturation/velocity arm is conditionally blocked, but the block is avoidable.** CytoTRACE2 (cytotrace2-py v1.1.0; Kang et al., *Nature Methods* 2025) runs on any expression matrix, is explicitly designed to suppress batch/platform variation, returns an absolute potency score in [0,1], and needs no spliced/unspliced data. RNA-velocity latent time (scVelo dynamical model) requires spliced/unspliced layers, which are *not* posted pre-computed for these studies and would require re-quantification from BAM/FASTQ via velocyto/kallisto. **Decision: skip velocity in Phase 1.**
5. **Budget is not the binding constraint** given 2026 neo-cloud pricing (details below). Idle/debug GPU time, not training, is the real cost driver.

Details

Task 0 — Environment (executable today; ~$0)

Pin a reproducible stack. Recommended versions as of mid-2026:

- `python=3.10`, `scanpy>=1.10`, `anndata>=0.10` (for `backed='r'` and on-disk concatenation)
- `scvi-tools>=1.2.2,<1.4` — provides `scvi.model.SCVI`, `scvi.model.SCANVI`, `scvi.external.ContrastiveVI`, and `scvi.train._callbacks.SaveCheckpoint`
- `scarches` (latest) for scPoli/surgery convenience, though scvi-tools native surgery is sufficient
- `cytotrace2-py==1.1.0` (PyPI / bioconda `cytotrace2-python`)
- `celloracle==0.20.0`
- `geosketch` (latest PyPI; `from geosketch import gs`)
- `POT>=0.9.6` (sliced-Wasserstein, debiased Sinkhorn) and `giotto-tda==0.5.1` (Vietoris-Rips persistence) or `gudhi`; consider `geomloss` for GPU-free Sinkhorn divergences
- `hnoca` (PyPI; HNOCA-tools `snapseed` + `map.AtlasMapper`) for label-transfer convenience
- `scvelo>=0.3` and optionally `dynamo-release` — install but treat as conditional/out-of-scope
  Use a single conda env exported to `environment.yml`; build a Docker image so the cloud pod boots identically.

Task 1 — Source and reconcile true raw UMI counts (HIGHEST CONSEQUENCE; partially blocked)

Step 1a (today, local/CPU; ~$0). Download and *inspect* the existing HNOCA objects before paying for anything. For each h5ad, check `adata.X.dtype` and `adata.X[:50].data` for integers, and enumerate `adata.layers.keys()` and `adata.raw`. Start with the full `hnoca_allmeta.h5ad` (zenodo 14160929) — described verbatim as containing "additional data representations and metadata from intermediate processing steps of the HNOCA," it is the object most likely to harbor a counts layer the pilot never used. The cleaned object is described as having "some metadata and representations originating from intermediate processing steps removed," so it is *less* likely to retain counts. **Pass criterion:** a layer whose values are non-negative integers with per-cell sums equal to `n_counts`. If found, the blocker collapses to a re-mapping job and Task 1 is essentially solved at zero data-acquisition cost.

Step 1b (today, CPU; ~$0). Pull the CELLxGENE Census / Discover HNOCA via `cellxgene_census.get_anndata(census, organism="Homo sapiens", X_name="raw", obs_value_filter=...)`. The CELLxGENE/Census schema mandates raw counts (Census stores "raw counts in X['raw']"), so this yields genuine integers — but the hosted object is ~216k cells (a subsample), covering only part of the 1.77M query. Use it to (i) validate that a raw-counts pipeline runs end-to-end and (ii) recover counts for whatever fraction of query cells are present, matched by barcode.

Step 1c (the robust path; per-study GEO/ArrayExpress). For the four Gate-C protocols, retrieve original raw matrices and reconcile to HNOCA cells:

- **Pellegrini 2020 (choroid plexus, 10x 3'v3):** ArrayExpress/BioStudies **E-MTAB-10500** (confirmed).
- **Treutlein 2023 (10x multiome):** = Fleck et al., *Nature* 621:365–372 (2023), "Inferring and perturbing cell fate regulomes in human brain organoids" (DOI 10.1038/s41586-022-05279-8; online 5 Oct 2022). 10x Chromium Single Cell Multiome ATAC+GEX, generated from 4-month-old organoids; GEO accession to be read from the paper's Data Availability section. Nature +2
- **He 2022 (10x 3'v3):** GEO accession to be read from HNOCA Supplementary Table 1.
- **Fiorenzano 2021 (midbrain DA, 10x 3'v3):** *Nat. Commun.* 12:7302 (DOI 10.1038/s41467-021-27464-5; note Author Correction Nat Commun 13:3312, 2022); GEO accession from the paper's Data Availability section / HNOCA Supplementary Table 1.

HNOCA Supplementary Table 1 (in the Nature paper) is the single consolidated source of per-dataset accessions and should be the first stop before any download.

Barcode reconciliation strategy. HNOCA `obs` carries dataset/batch identifiers (`id`, biological-replicate, donor) and original barcodes; integration was done with scPoli using a concatenated batch covariate. Build a mapping key `{dataset_id}_{original_barcode}`, intersect with each GEO matrix's barcodes (strip 10x `-1` suffixes), and reconcile genes to the 4,000-HVG space **by Ensembl ID, not symbol**. **Validation:** correlate `log1p(raw_reconstructed)` against the existing HNOCA lognorm layer per cell (expect Spearman > 0.95); flag any dataset whose cell-recovery rate < 90%.

Failure-mode branch (counts unrecoverable for some studies): proceed with whatever subset has genuine counts for the *confirmatory* analysis (the four Gate-C studies are the priority), and explicitly down-weight or exclude count-less studies from the ZINB surgery, documenting them as projected-only.

Spliced/unspliced: none of the four are expected to post spliced/unspliced layers; they are absent from GEO filtered matrices and would require velocyto/kallisto re-quantification from BAM/FASTQ. **Decision: do NOT re-quantify in Phase 1** — use CytoTRACE2 as the primary maturation axis; treat velocity as a Phase-2 stretch goal.

Task 2 — Harden the classifier (today; GPU ~0.5–1 hr; ~$1–3)

Retrain scANVI from the existing scVI weights with checkpoint selection and a non-uniform prior:

```python
from scvi.train._callbacks import SaveCheckpoint
ckpt = SaveCheckpoint(monitor="elbo_validation", load_best_on_end=True)  # on_except
scanvi.train(
    max_epochs=100,
    early_stopping=True,
    early_stopping_monitor="elbo_validation",
    early_stopping_patience=15,
    check_val_every_n_epoch=1,
    callbacks=[ckpt],
)
```

Also monitor held-out classification accuracy. **Class-prior choice:** because organoid query composition differs from the primary reference (off-target mesenchyme is rare in primary, common in failed organoids), a uniform prior mis-calibrates posteriors toward absent classes, while a pure-query-empirical prior is circular (query composition is partly what we are measuring). Use a **lightly-regularized empirical-Bayes prior**: estimate class frequencies from the query, then shrink toward flat via Dirichlet smoothing (mixing weight $\alpha \approx 0.1$–0.3 toward uniform). **Pass criteria:** held-out reference macro-F1 ⩾ pilot baseline; the pilot's over-training signature (val ELBO degraded ~15 from minimum) eliminated by best-checkpoint selection.

Task 3 — contrastiveVI dish-stripping with Control 3 (centerpiece)

3a. Configuration. Background = primary on-target cells; target = organoid on-target cells. The salient latent is the in-vitro/dish axis (the disentangled z_iv the pilot's mean-difference proxy only approximated).

```python
cvi = scvi.external.ContrastiveVI(adata, n_background_latent=30, n_salient_latent=10
                                  use_observed_lib_size=False)
cvi.train(background_indices=bg_idx, target_indices=tg_idx,
          early_stopping=True, max_epochs=500)
salient = cvi.get_latent_representation(adata, representation_kind="salient")
shared  = cvi.get_latent_representation(adata, representation_kind="background")
```

Requires raw counts (depends on Task 1 for the organoid side; primary reference already has counts). The adversarial subspace-separation objective is demoted to a verification check, not the primary mechanism.

3b. Disentanglement validation (pass/fail). (i) A classifier on the **salient** latent must predict organoid-vs-primary origin well (AUROC > 0.85). (ii) A classifier on the **shared** latent must *fail* to predict origin (AUROC ≈ 0.5–0.6). If shared still predicts origin, disentanglement leaked: increase `n_salient_latent`, raise the salient KL weight, or check for residual batch confounding before trusting any dish-stripped distance.

3c. Single vs. conditional dish axis. The literature predicts glycolytic/hypoxic shifts are pan-organoid while ER/UPR stress is lineage-specific. Test whether one salient subspace suffices: regress salient dimensions on cell-type/lineage identity; if lineage explains substantial salient variance (e.g. >25% by PERMANOVA), move to a **conditional/lineage-modulated dish axis** (condition salient inference on coarse lineage, or fit per-lineage salient offsets). Report this as a finding either way — it directly tests the pan-organoid-vs-lineage-specific stress hypothesis.

3d. Control 3 — maturation-matched baseline (what makes 0.906 interpretable).

- **Maturation scoring (lineage-independent, dish-robust):** run CytoTRACE2 (`cytotrace2-py` v1.1.0) on organoid and primary cells, yielding absolute potency scores. (If spliced/unspliced ever become available, cross-check with scVelo latent time on concordant-rank cells only — not required for Phase 1.)
- **Matched subsampling:** for each off-target/on-target comparison, stratified-resample the primary baseline into CytoTRACE2 potency bins so the matched-primary set has the same maturation histogram as the organoid set.
- **Statistical test:** in the shared latent, compute **sliced-Wasserstein** distance (POT `sliced_wasserstein_distance`, `n_projections ≥ 200`) between the organoid-convergent state and (a) the *matched* primary baseline and (b) the *unmatched* primary baseline. Cross-check with debiased Sinkhorn divergence (POT `empirical_sinkhorn_divergence`). Build the null by **permutation** (shuffle organoid/primary labels within maturation bins, ⩾1,000 permutations).
- **Interpretation gate:** convergence is "real" (not mere immaturity) only if the organoid-to-matched-primary distance stays significantly below a cross-lineage divergence scale *and* the dish-salient signal survives maturation matching. **Failure-mode branch:** if matching collapses the signal (matched distance ≈ self-distance floor), convergence is fully explained by shared immaturity — report this and pivot the narrative from "default state" to "arrested-immature state," which is itself a publishable result.

Task 4 — Interpretation and scale

4a. CellOracle levers (hypotheses, not interventions). Build/borrow a neural base-GRN, fit `celloracle==0.20.0` on the convergent off-target population, and run in-silico TF perturbations predicting shifts back toward on-target identity (the package's signal-propagation simulation on the GRN coefficient matrix). Frame outputs explicitly as candidate hypotheses for wet-lab validation. GRN inference is CPU/memory-bound; keep off GPU.

4b. In-vivo program anchoring. Score the convergent state against curated developmental signatures (primitive streak, early/paraxial mesenchyme, neural crest, EMT) via `scanpy.tl.score_genes`. **Pass criterion for a "default state" claim:** the convergent population scores significantly higher on a coherent early-mesenchymal/primitive program than on its lineages-of-origin programs.

4c. Geometry & statistics (CPU-only; carry forward the project's design):

- **Primary metric:** sliced-Wasserstein (POT) — n^(−1/2) convergence independent of dimension. **Cross-check:** debiased Sinkhorn divergence (POT, Feydy et al. 2019 construction). **Never** raw Wasserstein in 30-D latent space.
- **Interpret every distance only relative to** a matched-N self-distance floor (split-half of the same population) and the Control-3 baseline.
- **Minimum stable N by rarefaction:** subsample ladder (100/200/…/2000 cells), plot split-half self-distance vs N, fix N* where it saturates.
- **Topology:** H0 persistent homology (giotto-tda `VietorisRipsPersistence`) with Fasy-style bootstrap confidence bands (subsample B bootstraps, average via persistence landscapes per Chazal/Fasy et al.) as confirmatory; H1 exploratory only.
- **Power:** simulation-based (no closed form for a permutation test on sliced-Wasserstein) — simulate convergent vs non-convergent mixtures at the fixed N* and estimate detection power.

4d. Gate-C concentration guard (Treutlein 2023 = 4,670 cells = 61.7% of off-target mesenchyme). Mandatory: **leave-one-study-out (LOSO)** re-computation of every convergence statistic, plus **per-study down-weighting** (weight cells inversely to study size) for the pooled estimate. **Restrict confirmatory claims to the four studies each clearing 300 cells** (Treutlein 4,670; He 782; Pellegrini 549; Fiorenzano 309; the 47 sub-300 protocols totaling 1,258 cells are exploratory only). **Failure-mode branch:** if dropping Treutlein 2023 abolishes the signal (LOSO distance jumps outside the CI of the all-studies estimate), the signal is a single-study artifact — downgrade to "observed in the Treutlein multiome dataset, not yet generalizable," and make broad protocol stratification an explicit Phase-2 acquisition goal.

Task 5 — Build for scale (today; mostly CPU)

- **geosketch** to preserve rare off-target populations: run `gs(X_latent, N)` on the existing scVI latent so dense on-target regions are subsampled aggressively while rare off-target mesenchyme is retained (Hie et al. 2019 explicitly show geometric sketches capture rare cell types with high sensitivity; uniform downsampling would erase the very cells under study). Build training subsets of ~150–400k cells.
- **Lean I/O:** `anndata` backed mode (`backed='r'`) and on-disk concatenation; never densify the full 1.77M-cell matrix. The pilot's 251 GB container ceiling is then comfortably respected.
- **Mixed precision** for all scvi-tools training (`precision="16-mixed"`).
- **Sequence the heavy frameworks:** contrastiveVI first, then CellOracle — never concurrently; the geometry/topology stack runs on CPU in between.

GPU-hours and dollar costs (against USD 150–200)

2026 neo-cloud single-GPU pricing (verified ranges): RunPod H100 on-demand ~$1.99–2.39/hr (Community vs Secure); Thunder Compute H100 $1.38/hr; Spheron H100 ~$2.01/hr on-demand, spot toward ~$1.03/hr; A100 80GB ~$1.07–1.40/hr; L40S $0.19–0.86/hr; H100 spot as low as ~$0.50/hr on marketplaces. Billing is per-second/minute. Spheron

Step	Device	Est. GPU-hr	Est. $ (~$2/hr H100, w/ margin)
scVI re-train on geosketch subset (only if HVG/genes change)	H100/A100	1.5–2.5	$3–5
scANVI hardening (Task 2)	H100/A100	0.5–1	$1–2
scArches surgery w/ real counts (partial retrain on subset + project rest)	H100/A100	1.5–3	$3–6
contrastiveVI (train once on on-target subset)	H100/A100	2–4	$4–8
CytoTRACE2	L40S/CPU-GPU	0.5–1	$1–2
CellOracle	CPU	0 (GPU)	$0
Geometry/topology (POT, giotto-tda)	CPU	0 (GPU)	$0
Subtotal compute		**~7.5–14.5 GPU-hr**	**~$12–23**
Re-runs / debugging buffer (×2.5)			~$30–55
Storage + egress (≈0.5–1 TB, network volume @ ~$0.07/GB/mo)			~$20–40
Total			**~$70–120, within $150–200**

Budget-risk flags: (1) If raw counts must be re-quantified from BAM/FASTQ, CPU-hours and storage balloon — *avoid by using CytoTRACE2 and per-study filtered matrices*. (2) contrastiveVI on the full 1.77M cells would blow both memory and budget — *always train on the geosketch subset and project*. (3) Idle GPU time while debugging is the largest real cost — *develop on CPU or a cheap L40S, reserve H100 only for final training runs*, and use spot instances with `SaveCheckpoint(on_exception=True)` so a preemption resumes from the best checkpoint.

Recommendations

Start today (no new data needed):

1. Pin the environment (Task 0) and build the Docker image.
2. Download and inspect all three HNOCA Zenodo objects for a hidden counts layer (Task 1a) — this one cheap step may dissolve the main blocker.
3. Harden the classifier (Task 2) on the existing latent space.
4. Stand up the CPU geometry/statistics stack (Task 4c), fix N* by rarefaction, and pre-register the permutation/power design.
5. Run CytoTRACE2 and build the Control-3 matched-subsampling machinery (Task 3d).

Then, gated on raw counts: 6. If Task 1a/1b/1c yields integer counts for the Gate-C studies, run genuine scArches surgery and re-classify; expect the ambiguous_novel fraction (37.7%) to drop and true_offtarget (17.8%) to rise as real off-target cells are pulled off the soft boundary. 7. Train contrastiveVI on the on-target subset, validate disentanglement (3b), test single-vs-conditional dish axis (3c), run Control 3 (3d). 8. CellOracle levers (4a) + signature anchoring (4b), with the Gate-C concentration guard (4d) applied to every confirmatory statistic.

Decision thresholds that change the plan:

- *Counts unrecoverable for >1 Gate-C study* → restrict confirmatory claims to studies with genuine counts; flag the rest projected-only.
- *Control 3 collapses the signal* → reframe as arrested-immaturity (still publishable).
- *LOSO without Treutlein abolishes convergence* → downgrade to single-dataset observation; make protocol-stratified acquisition the Phase-2 priority.
- *contrastiveVI shared latent still predicts origin (AUROC > ~0.65)* → iterate salient dimensionality/KL weight before trusting any dish-stripped distance.

FINAL VERDICT

Executable immediately, with data already in hand (start today):

- Task 0 environment build; Task 1a/1b inspection of HNOCA Zenodo objects and CELLxGENE pull; Task 2 classifier hardening (runs on existing scVI weights + reference counts); Task 3d CytoTRACE2 maturation scoring and Control-3 matched-subsampling scaffolding; Task 4b signature anchoring; Task 4c the entire CPU geometry/statistics/topology stack and rarefaction; Task 5 geosketch + backed-I/O scale engineering. CellOracle (4a) can be stood up on the existing latent now and re-run once counts arrive.

Blocked pending data acquisition:

- **Genuine scArches surgery and re-classification (Task 1c → 6)** and **contrastiveVI on the organoid side (Task 3a–3c)** both require true raw integer UMI counts for the organoid query. These are *high-probability recoverable* (CELLxGENE raw layer is schema-guaranteed; Pellegrini = E-MTAB-10500; the other three accessions live in HNOCA Supplementary Table 1 and each paper's Data Availability section) but are *not yet confirmed in hand*. This is the gating dependency for the headline result.
- **RNA-velocity latent time** is hard-blocked (no pre-computed spliced/unspliced; re-quantification deliberately out of scope). CytoTRACE2 fully substitutes for Phase 1.

Beneficial but NOT strictly required (no expert/PI reply needed to proceed):

- Confirmation from the HNOCA authors of which object the pilot used and whether the lab can share per-dataset raw counts directly would *accelerate* Task 1 but is not required — every task can proceed from public data. PI input on the developmental-program signature panels (4b) and on whether single-vs-conditional dish modeling (3c) matches their biological priors would *improve* interpretation but does not gate execution.

Bottom line: the researcher can begin productive work today on more than half of Phase 1, the entire enterprise fits the budget several times over, and the one true blocker — raw UMI counts — is a verification-and-download task with at least three independent recovery routes rather than a genuine dead end.

Caveats

- **No HNOCA Zenodo object is confirmed to contain raw integer counts**; step 1 is a verification, and the CELLxGENE-hosted version is a ~216k-cell subsample, not the full 1.77M query.
- **Three of four Gate-C GEO accessions were not pinned** in this research pass; HNOCA Supplementary Table 1 is the authoritative consolidated source. Only Pellegrini's ArrayExpress accession (E-MTAB-10500) is confirmed here; the Treutlein/Fleck multiome study is identified (Fleck et al., Nature 621:365–372, 2023) but its GEO number must be read from the paper.
- **Spliced/unspliced layers do not exist pre-computed** for these studies; RNA-velocity maturation is therefore out of scope for Phase 1.
- **Cloud prices fluctuate** and spot availability is not guaranteed; dollar estimates carry a deliberate ×2.5 debugging buffer and assume neo-cloud (RunPod/Vast/Spheron/Thunder), not hyperscaler, rates.
- This plan deliberately does **not** add kidney/endoderm systems — that is Phase 2 — and treats CellOracle levers as candidate hypotheses, never validated interventions.